



**S. B. JAIN INSTITUTE OF
TECHNOLOGY, MANAGEMENT &
RESEARCH, NAGPUR**

Practical No. 12 (Postlab)

Aim: To study and implement artistic image transformation using the AI-based tool **DeepArt**, and understand how neural networks can recreate images in the style of famous artworks.

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AIM: To study and implement artistic image transformation using the AI-based tool **DeepArt**, and understand how neural networks can recreate images in the style of famous artworks.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

- To understand the concept of neural style transfer
- To explore AI-based artistic image generation
- To learn how DeepArt transforms images using neural networks
- To analyze the output images and compare them with original inputs

THEORY:

DeepArt is an AI-based image processing tool that uses Neural Style Transfer (NST) to convert ordinary images into artistic masterpieces. This technique is a practical application of Deep Learning, which is a subset of artificial intelligence.

Neural Style Transfer (NST)

Neural Style Transfer is a method that blends two images:

- Content Image → Defines the structure (objects, layout)
- Style Image → Defines artistic features (colors, brush strokes, texture)

The goal is to generate a new image that maintains the content of the first image while adopting the artistic style of the second.

Role of Convolutional Neural Networks (CNNs)

DeepArt uses **Convolutional Neural Networks (CNNs)**, which are specially designed for image processing tasks.

A CNN performs:

- Feature Extraction: Detects edges, shapes, textures
- Layer-wise Learning:
 - Lower layers → detect simple patterns (edges, lines)
 - Higher layers → detect complex structures (objects, faces)

Style Representation

The style is extracted using a mathematical concept called the **Gram Matrix**:

- It captures correlations between different feature maps
- Represents textures, colors, and patterns of the style image

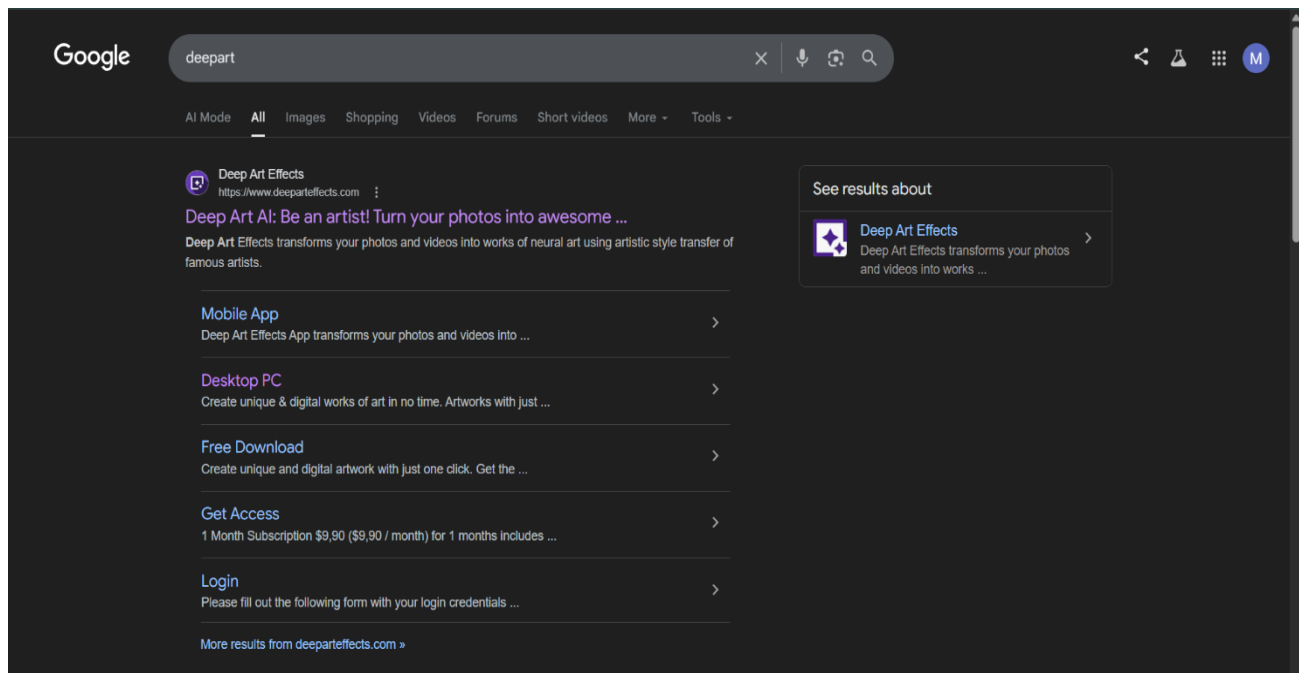
This allows the system to mimic:

- Brush strokes
- Color distribution
- Artistic patterns

Procedure / Steps

1. Open the DeepArt website.
2. Upload a content image (your original photo).
3. Upload or select a style image (e.g., famous artwork).
4. Choose the desired resolution or quality settings.
5. Click on Generate / Create Artwork.
6. Wait for the AI model to process the images.
7. Download the generated stylized image.
8. Compare the output with the original images.

INPUT & OUTPUT:



The image shows a screenshot of the Deep Art AI website and its application interface. The website has a purple header with the logo and navigation links: Deep Art Creator, Deep Art Effects, For Business, and Sign In. The main content area features a large image of a woman's profile with the text: "CREATIVE INTELLIGENCE – THE FUTURE OF IMAGING. LOCAL. NO CLOUD. PRIVACY IS OUR PRIORITY". Below this, it says: "With Deep Art Creator and generative intelligence, you can turn your ideas into breathtaking pictures. Deep Art Effects allows you to easily and innovatively edit images using artificial intelligence. Simple, innovative and offline." A "TRY NOW" button is present. A "KNOWN FROM The New York Times" badge is also visible. Below the main content, there's a section titled "POWERFUL IMAGE PROCESSING WITH ONE CLICK THROUGH ARTIFICIAL INTELLIGENCE" showing a grid of eight artistic images. At the bottom of the website section, a small text reads: "Create unique works of art from your pictures. Scale images to four times the original size without loss of quality. Optimize or color images automatically, imitate expensive SLR cameras, automatically cut out backgrounds. All this offline and the pictures stay with you! With Deep Art". Below the website screenshot is the application interface. It has a dark theme with a menu on the left containing "Text To Image" and "Image To Image". A central area shows a "Preparing AI..." status. The right panel is titled "Image To Image" and includes a "Prompt" input field, a "Predefined Styles" button, an "Input Image" section with a landscape image, a "50% Image strength" slider, and "Start" and "Stop" buttons.

Deep Art AI

Deep Art Creator Deep Art Effects For Business Sign In

CREATIVE INTELLIGENCE – THE FUTURE OF IMAGING. LOCAL. NO CLOUD. PRIVACY IS OUR PRIORITY

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Deep Art Creator

File Info

Text To Image Image To Image

Preparing AI...

Image To Image

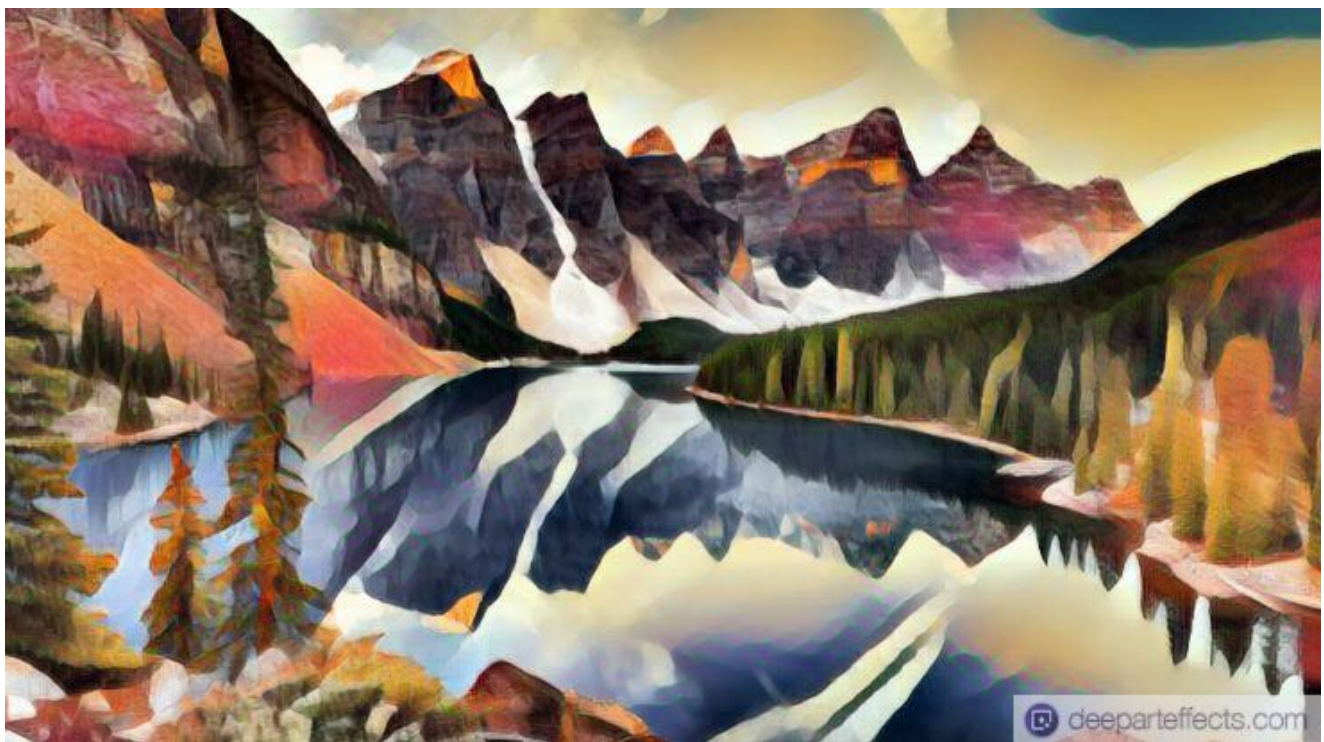
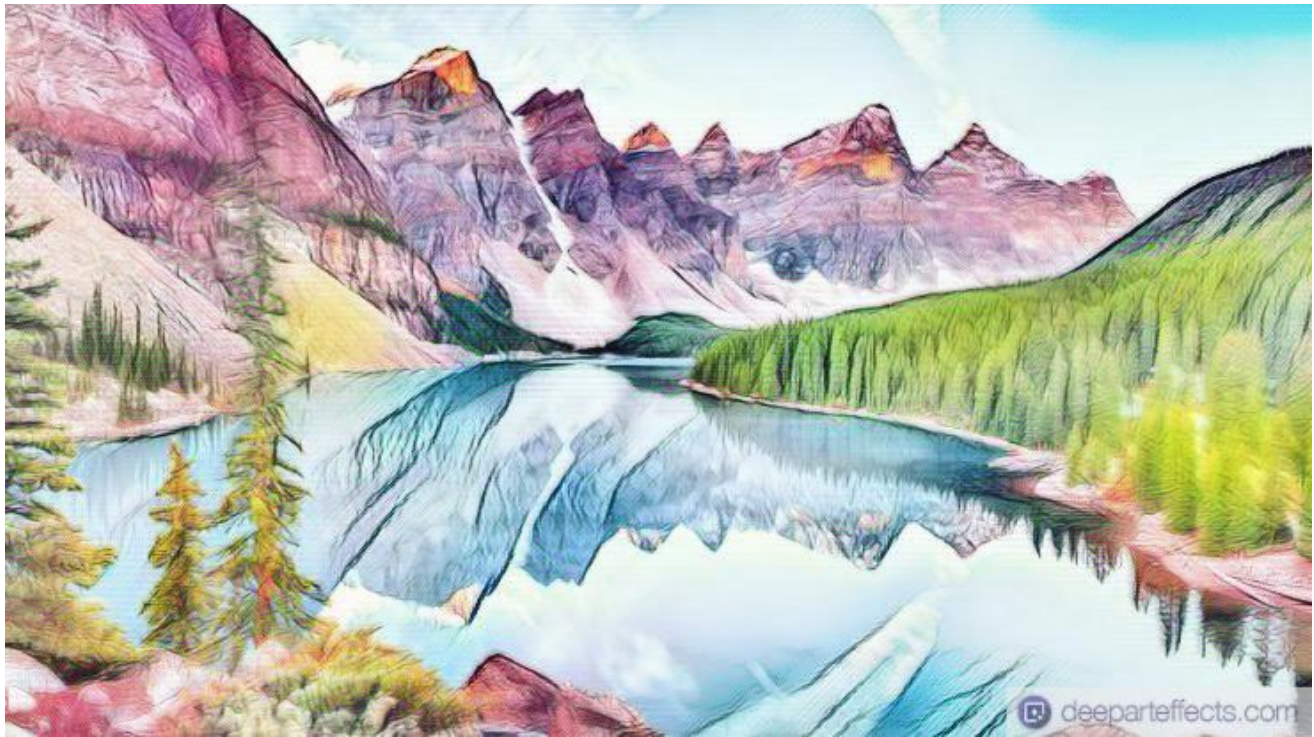
Prompt

Predefined Styles

Input Image

50% Image strength

Start Stop



CONCLUSION: AI image tools like Deep Art Creator allow easy transformation of images into various artistic styles. Prompt clarity and parameter tuning are crucial for achieving desired results.

DISCUSSION QUESTIONS:

1. What does the Image-to-Image mode do in Deep Art Creator?

2. Why do we need to write a prompt for the AI?

3. How does changing the image strength affect the result?

REFERENCES:

- https://www.deeparteffects.com/?srsltid=AfmBOopf1XKxsx9EftROAjdUyOMg9BFCM_xR-aXkbNdkNlb750MRljfK
- <https://github.com/Deepart-Ai>
- <https://portraitart.app/blog/deepart>